

Day 4 : GPUs and Cooling Systems Theory

Task 1 : Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

GPU 0 – Integrated GPU

- **Brand:** Intel
- **Model:** Intel(R) UHD Graphics
- **Type:** Integrated GPU
- **Purpose:** Handles normal tasks such as Windows, web browsing, videos, and other light graphics work.

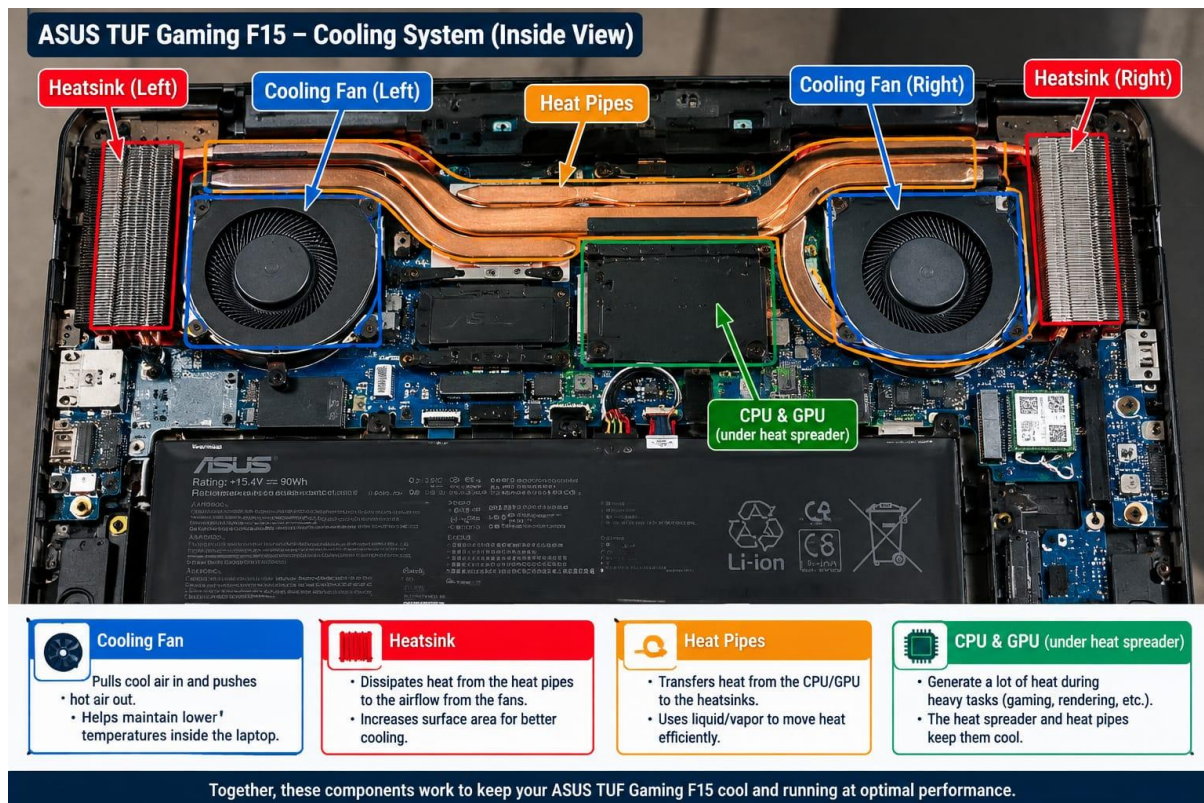
GPU 1 – Dedicated GPU

- **Brand:** NVIDIA
- **Model:** NVIDIA GeForce RTX 3050 Laptop GPU
- **Type:** Dedicated GPU
- **Dedicated Memory:** 4 GB
- **Purpose:** Provides higher graphics performance for gaming, video editing, 3D applications, etc.

Short Conclusion

My ASUS TUF Gaming F15 has **two GPUs**: an **Intel UHD Graphics integrated GPU** and an **NVIDIA GeForce RTX 3050 Laptop GPU dedicated GPU**. The NVIDIA RTX 3050 is the main high-performance GPU for demanding games and graphics applications.

Task 2 : Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.



Cooling System – ASUS TUF Gaming F15

The cooling system inside the ASUS TUF Gaming F15 consists of cooling fans, heatsinks, and heat pipes. These components work together to remove heat generated by the CPU and GPU and maintain safe operating temperatures.

- **Cooling Fan:** Pulls cool air in and pushes hot air out of the laptop.
- **Heatsink:** Dissipates heat transferred from the heat pipes and increases the surface area for cooling.

- **Heat Pipes:** Transfer heat away from the CPU/GPU to the heatsinks.
- **CPU & GPU:** Generate heat during demanding activities such as gaming and rendering. The cooling system helps prevent overheating.
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Conclusion:

Together, these components keep the ASUS TUF Gaming F15 cool and help maintain stable performance.

Task 3 : Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Air Cooling vs Liquid Cooling for GPUs

Factor	Air Cooling	Liquid Cooling
Cooling method	Uses a fan and heatsink to remove heat from the GPU.	Uses coolant, water block, pump, and radiator to remove heat.
Cost	Generally cheaper and affordable.	Generally more expensive because of additional components.
Efficiency	Provides good cooling for normal gaming and everyday use.	Provides better cooling for high-performance gaming and heavy workloads.
Maintenance	Easy to maintain ; mainly requires cleaning dust from fans and heatsinks.	Requires more maintenance , such as checking the pump, tubes, and coolant.
Installation	Simple and easy to install.	More complicated to install.
Noise	Fans can become noisy at high speeds .	Can be quieter in some systems , but radiator fans and pumps still make noise.
Risk	No risk of liquid leakage.	Has a small risk of coolant leakage if the system fails.

Task 4 : Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

Air Cooling vs Liquid Cooling

I would choose Liquid Cooling for the GPU.

1. **Better heat management:** Liquid cooling can remove heat more efficiently during heavy gaming and streaming.
2. **Graphics-heavy games:** High-end games put more load on the GPU and generate more heat, so liquid cooling is a good option.
3. **Long gaming sessions:** It helps maintain a stable temperature when the PC is running for several hours.
4. **Better performance:** Good temperature control can reduce overheating and thermal throttling, helping the GPU maintain performance.
5. **Cost and complexity:** Liquid cooling is more expensive and harder to install than air cooling.
6. **Conclusion:** For a **high-performance gaming PC used for IPL streaming and graphics-heavy games**, I would choose **liquid cooling** because of its better heat management.